HAND PARALYSIS REHABILITATION SYSTEM

REPORT

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1. ABSTRACT

Hand paralysis is one of the most disabling outcomes of neurological injury, often resulting from stroke, nerve damage, or spinal cord injury. It severely restricts a person's ability to perform basic daily activities such as grasping, lifting, and writing. Conventional rehabilitation systems primarily target finger actuation, offering simple opening and closing exercises for the hand. However, these solutions fail to provide complete hand recovery as they neglect the training of the wrist joint, which plays a vital role in restoring full functional movement.

To overcome this limitation, the **Hand Paralysis Rehabilitation System** has been designed as a smart, electromechanical rehabilitation device that focuses on both **finger** and **wrist rehabilitation**. The system supports three primary types of motion: **finger folding/unfolding**, **wrist flexion/extension**, and **wrist ulnar/radial deviation**. These exercises collectively train multiple muscle groups and improve coordination between the fingers, wrist, and forearm, leading to more effective recovery.

The system operates in two modes — Training (machine-assisted) and Testing (patient-controlled) — allowing progressive rehabilitation from assisted to voluntary motion. Sensor data from each session are uploaded to the cloud for analysis (see Section 6 – Operational Phases).

The system's intelligent design ensures smooth motion control, precise data acquisition, and secure cloud storage. Unlike traditional rehabilitation systems that are bulky, costly, or limited in range of motion, this project offers a **low-cost, portable, and multi-axis rehabilitation platform**. It bridges the gap between mechanical physiotherapy and digital health monitoring, providing both active assistance and passive assessment for comprehensive hand rehabilitation.

With its ability to perform complex multi-joint motions, record real-time motion data, and offer personalized rehabilitation routines, the Hand Paralysis Rehabilitation System represents a significant step toward smarter, more accessible, and data-driven recovery solutions for patients with hand paralysis.

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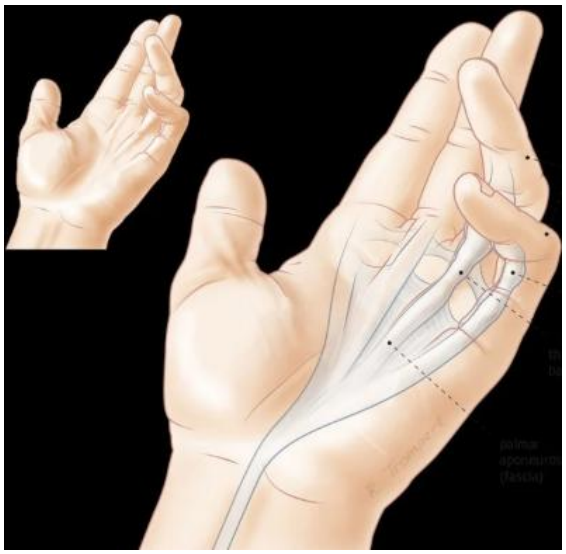
2. INTRODUCTION

The human hand is an intricate biomechanical structure capable of performing precise, coordinated movements through the combined function of the fingers, wrist, and forearm. In cases of paralysis caused by stroke, peripheral nerve injury, or muscular dysfunction, the neuromuscular connection between the brain and these motor units becomes impaired. Rehabilitation aims to restore this lost connection through repeated, guided motion exercises that stimulate muscle memory and re-train motor neurons.

However, traditional therapy sessions require constant supervision by a physiotherapist and are often limited by time, cost, and availability. Mechanical rehabilitation devices exist, but most commercially available systems are designed to train **finger grasping motions** only, ignoring the complex rotational and bending actions of the **wrist**. Complete recovery requires training across all relevant axes of motion, particularly **flexion-extension** and **ulnar-radial deviation**.

The **Hand Paralysis Rehabilitation System** was conceived to address these gaps. It offers a mechatronic solution capable of executing multi-axis movements with precise control while collecting motion data for performance evaluation. By combining **embedded systems**, **actuators**, and **cloud computing**, the project delivers both the mechanical assistance and analytical insight necessary for efficient and measurable rehabilitation.



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3. PROBLEM STATEMENT

Patients suffering from hand paralysis experience significant difficulty in performing everyday tasks due to loss of voluntary muscle control. Existing rehabilitation machines predominantly focus on finger motion-specifically, the opening and closing of the hand. These devices fail to train the **wrist joint**, which is essential for a full range of hand movements. Without proper wrist mobility training, patients struggle with essential actions like holding, rotating, or positioning objects.

Therefore, there is a clear need for a **low-cost, multi-axis rehabilitation device** that assists in both **finger movement** and **wrist control**, enabling more comprehensive therapy and faster recovery. This system should also provide **quantitative progress data** to therapists for analysis and adaptive training plans.

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4. EXISTING REHABILITATION SYSTEMS

Before developing the proposed system, an extensive review of existing hand rehabilitation devices was conducted. Most commercial and research-based solutions currently available on the market were analyzed in terms of **functionality, cost, motion range, and data acquisition capability**. While these systems have contributed to the field of rehabilitation engineering, they exhibit several **technical and practical limitations** that restrict their accessibility and effectiveness.

4.1 Existing Products and Techniques

1. **MIT Manus (InMotion2) Robotic Arm – USA**
   - A large, clinic-based robotic arm designed for stroke rehabilitation.
   - Supports repetitive upper limb exercises with adaptive resistance.
   - **Limitations:** High cost (>$20,000), large size, limited availability in smaller clinics, and lacks portability for home therapy.
2. **HandTutor Glove – MediTouch (Israel)**
   - Uses motion sensors in a fabric glove to track finger and wrist movements.
   - Offers visual biofeedback for hand exercises.
   - **Limitations:** Focuses mainly on flexion–extension; no active actuation for assisted movement; dependent on therapist supervision.
3. **SaeboGlove – Saebo Inc. (USA)**
   - A spring-assisted mechanical glove that helps patients open and close their fingers.
   - **Limitations:** No data collection or digital feedback; provides passive motion assistance only; lacks wrist joint training.
4. **ExoHand – Festo Robotics (Germany)**
   - Pneumatic exoskeleton with precise movement replication.
   - **Limitations:** Highly complex and expensive; not suitable for individual clinical or home-based use; requires external air compressor and power setup.
5. **DIY/Academic Research Prototypes (Various)**
   - Many research projects focus on 3D-printed robotic hands controlled via servo motors or EMG signals.
   - **Limitations:** Typically limited to single motion type, lack integrated data tracking, and are not medically validated.

4.2 Common Limitations in Existing Devices

From the analysis of current rehabilitation solutions, the following limitations were identified:

- **Limited Motion Range:** Most systems focus solely on finger flexion and extension, ignoring wrist mobility (flexion, extension, and deviation).
- **Lack of Real-Time Feedback:** Few devices provide quantitative data such as angle, speed, or repetition count to track improvement.
- **High Cost and Complexity:** Medical-grade robotic systems are expensive and non-portable, restricting access to hospital environments.
- **Absence of Cloud Connectivity:** Most systems lack IoT integration for data storage or remote monitoring.
- **Dependence on Therapist Supervision:** Patients cannot perform therapy independently or review their progress without professional guidance.
- **No Adaptive or Predictive Intelligence:** Existing solutions do not analyze performance data to predict or personalize recovery trajectories.

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5. PROPOSED SOLUTION

To overcome the above challenges, the **Hand Paralysis Rehabilitation System** has been designed as a **compact, intelligent, and multi-axis rehabilitation platform** that combines **mechanical actuation**, **sensor-based feedback**, and **data analytics** into one unified device.

5.1 How the Proposed System Addresses Limitations:

Limitation in Existing Systems	**Proposed System Improvement**
Limited to finger motion	Trains both finger and wrist across three distinct modes: finger flexion/extension, wrist flexion/extension, and wrist ulnar/radial deviation.
No quantitative feedback	Integrates MPU6050 IMU sensor to measure real-time angular motion, repetitions, and range of motion (ROM).
No cloud data logging	Uses Firebase Realtime Database for remote data storage and visualization of rehabilitation progress.
High cost and bulky design	Built using low-cost, off-the-shelf components and 3D-printed parts, making it affordable and portable.
Requires constant therapist supervision	Enables home-based, self-guided therapy with automatic logging and monitoring features.
No predictive intelligence	Employs machine learning models (Linear, Polynomial, and LSTM Regression) to predict recovery trends and estimate therapy timelines.
Lack of wrist deviation training	Incorporates servo-driven ulnar–radial deviation for comprehensive wrist rehabilitation.

5.2 Unique Features of the Proposed System:

- Dual-Phase Operation: Training (machine-assisted) and Testing (patient-controlled).
- Real-Time Sensor Data Logging: Continuous IMU feedback uploaded to cloud.
- Data-Driven Analytics: Python-based analysis to visualize and forecast recovery trends.
- Compact and Portable Design: Lightweight and battery-operable, suitable for both clinical and home use.
- Therapist Integration: Enables remote monitoring and performance evaluation without the need for specialized hardware.

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6. OBJECTIVES

The objectives of this project are:

1. To design and implement a **smart rehabilitation system** that assists in restoring hand motion in paralyzed patients.
2. To develop a **dual-phase mechanism** consisting of a **Training Phase** (machine-assisted) and a **Testing Phase** (patient-controlled).
3. To support three major rehabilitation modes:
    - Finger folding and unfolding
    - Wrist flexion and extension
    - Wrist ulnar and radial deviation

4. To use **ESP32** as the central control unit for actuator management and cloud connectivity.
5. To capture and log real-time motion data such as angle, mode, and repetitions using **Firebase Realtime Database**.
6. To enable remote monitoring and analysis of rehabilitation performance through data logging and visualization.
7. To ensure the device is **safe, compact, and cost-effective** for both clinical and home-based use.

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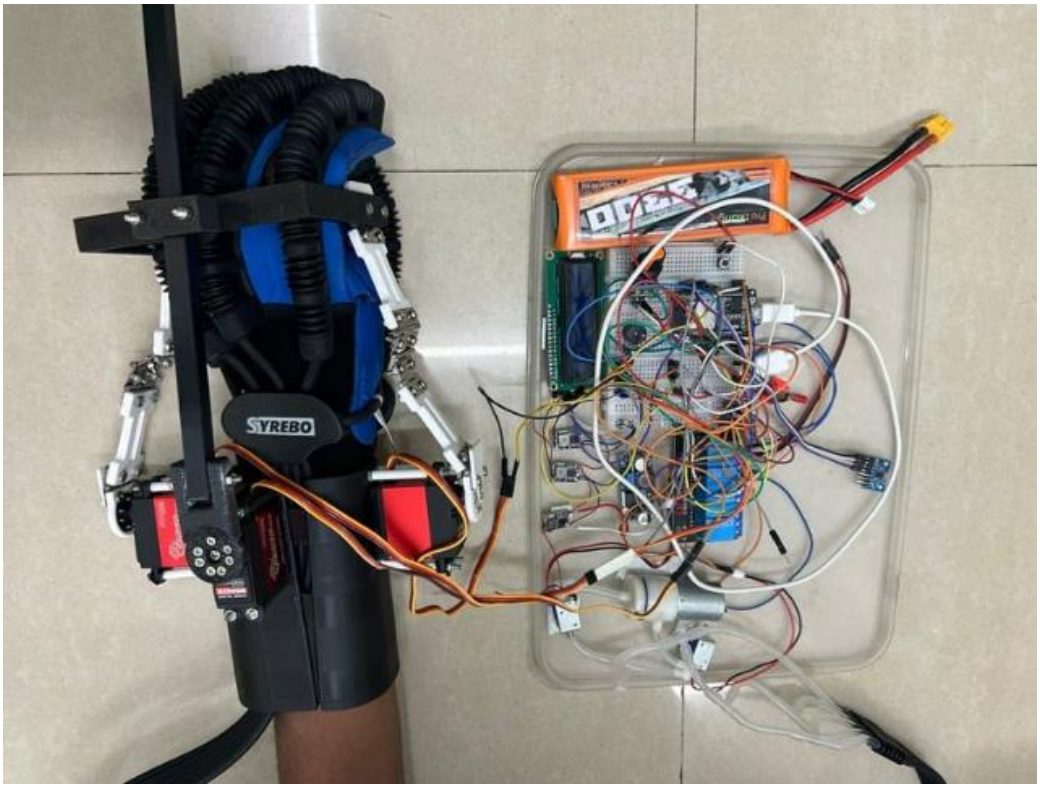
7. SYSTEM OVERVIEW

The Hand Paralysis Rehabilitation System operates as an integrated mechatronic platform combining **mechanical actuation**, **sensor feedback**, and **cloud-based data monitoring**.

Key Features

- Two operational phases: Training and Testing.
- Three selectable modes targeting distinct motion patterns.
- Real-time sensor feedback using an IMU.
- Wireless data logging and remote monitoring.
- Smooth, coordinated actuation using servo and pneumatic systems.

A toggle switch selects between **Training** and **Testing** phases, while a mode button chooses the motion type. Detailed phase behavior is provided in **Section 6**.



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8. OPERATIONAL PHASES

6.1 Training Phase

This phase provides **machine-guided movement** where the actuators perform controlled exercises for the patient. It allows the patient's muscles to experience repetitive, safe, and natural motion, helping restore neuromuscular coordination.

Mode 1: Finger Folding and Unfolding

- Uses a **pneumatic pump** connected to an inflatable air bladder under the glove's fingers.
- When inflated, the air bladder contracts the fingers into a fist; when deflated, it opens them.
- A **solenoid valve** controls airflow, synchronized with the pump.
- Helps rebuild finger flexor and extensor strength.

Mode 2: Wrist Flexion and Extension

- Utilizes **two servo motors** to control upward and downward wrist movements.
- Servo position values are varied smoothly to avoid jerky motion.
- The motion trains **forearm flexor** and **extensor** muscles, critical for lifting and lowering the wrist.

Mode 3: Wrist Ulnar and Radial Deviation

- Controls lateral (side-to-side) wrist motion.
- Strengthens **wrist stabilizer muscles** and improves control for complex hand postures.
- This mode is rarely addressed by traditional rehabilitation devices.

6.2 Testing Phase

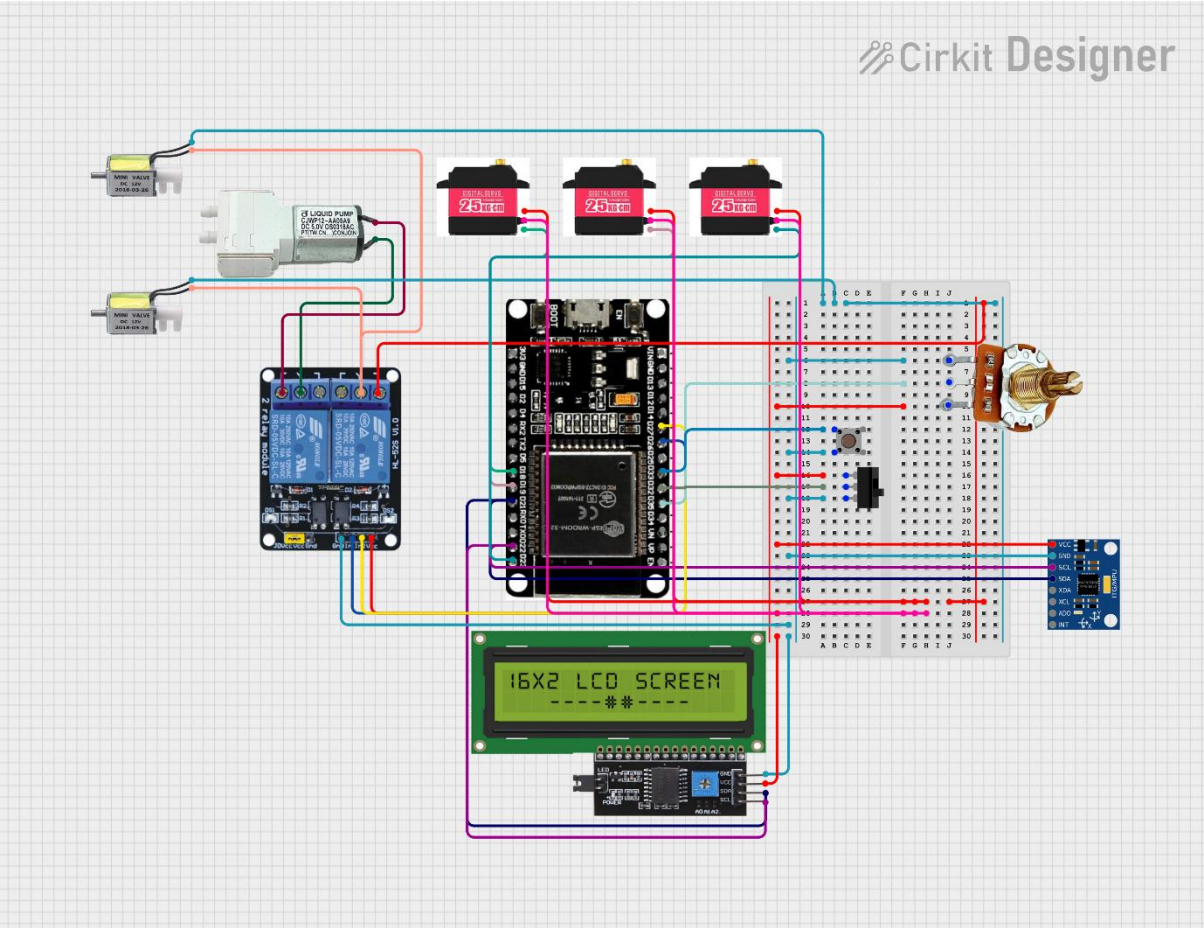
Once sufficient muscle activity is regained, the system is switched to Testing Phase. Here, the actuators are inactive; the patient moves their wrist or fingers voluntarily.

- The **MPU6050 IMU sensor** attached near the wrist records the **angular displacement**.
- Data parameters such as:
    - Session ID
    - Mode number
    - Repetition count
    - Angle value
    - Timestamp

    are uploaded to **Firebase Realtime Database** every second.
- This data allows physiotherapists to evaluate the range, accuracy, and improvement trends across multiple sessions.

9. HARDWARE SPECIFICATIONS AND DESIGN

Component	Specification	Function
ESP32 Microcontroller	Dual-core 32-bit MCU with Wi-Fi & BLE	Main control unit, handles input, actuator control, and Firebase communication
MPU6050 IMU Sensor	3-axis gyroscope + 3-axis accelerometer	Measures wrist angles and motion orientation
LCD Display	16 x 2 characters	Used to display information
Servo Motors (x3)	180° rotation, metal gear	Drive wrist flexion, extension, and ulnar/radial deviation
Pneumatic Pump	7V DC mini air pump	Provides pressurized air to fold fingers
Solenoid Valve	Normally closed, 12V DC	Regulates air flow for inflation/deflation cycles
Toggle Switch	ON-OFF configuration	Switches between Training and Testing phases
Mode Button	Tactile push button	Cycles through operational modes
Potentiometer	Analog Input	Used to set the number of repetitions
Power Supply	5V, 7V & 12V regulated	Powers ESP32, servos, and pneumatic components



10. SOFTWARE SPECIFICATIONS

The software architecture of the Hand Paralysis Rehabilitation System is designed as a hybrid framework combining embedded control (ESP32), cloud communication (Firebase), and data analytics (Python).
Each component plays a crucial role in ensuring synchronized actuation, reliable data transfer, and intelligent rehabilitation analysis.
The software consists of multiple integrated modules developed using Arduino IDE (for ESP32) and Python (for post-processing, data analysis, and visualization).

10.1 Embedded Control Software (Arduino IDE + ESP32)
The ESP32 microcontroller is programmed using the Arduino IDE, utilizing the following libraries:

- Firebase ESP32 Client for cloud communication,
- Wire.h and Adafruit_MPU6050.h for IMU data acquisition,
- Standard Servo and GPIO libraries for actuator control.

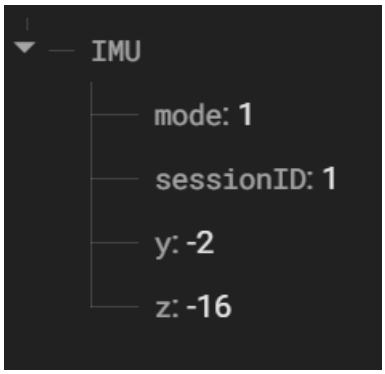
Initialization

- Establishes Wi-Fi connection using SSID and password credentials.
- Initializes the Firebase Realtime Database connection for data upload.
- Configures GPIO pins for toggle switch, mode selection button, servo control, and pneumatic pump activation.
- Calibrates the MPU6050 IMU for offset correction and stable angle readings.
- Displays system status on the LCD interface.

Control Logic
The firmware continuously monitors operational states:

- If Training Phase:
    - Executes predefined servo and pneumatic sequences based on the selected mode.
    - Controls actuator speed using PWM (Pulse Width Modulation) for smooth and natural motion.
    - Synchronizes the pump and solenoid valve operation for finger flexion and extension cycles.
- If Testing Phase:
    - Disables all actuators for safety.
    - Reads real-time IMU angle data (Y and Z axes) every second.
    - Formats the data as JSON and uploads it to Firebase Realtime Database.
    - Each log entry includes session ID, mode number, angular values, and timestamp.

Firebase Data Format:



Each record represents a time-stamped sensor measurement corresponding to a patient's wrist motion during the rehabilitation session.

10.2 Cloud Database and Data Synchronization (Firebase)

Firebase Realtime Database acts as the central cloud platform linking the ESP32 hardware and the Python-based analysis software.

- Real-time Uploads: The ESP32 transmits each data point via REST API calls.
- Secure Storage: Firebase authenticates each data write using unique database tokens, preventing unauthorized access.
- Scalable Architecture: Data is organized hierarchically by session IDs, enabling multi-session analysis and long-term trend tracking.
- Accessibility: Data can be retrieved directly by Python scripts for offline analytics or live dashboards.

Firebase enables continuous, wireless synchronization between patient device and analytical platform — eliminating manual data handling.

10.3 Data Analytics and Modeling (Python Environment)

Once session data is stored in Firebase, it is exported as a CSV file and processed in Python. The analytical system uses a combination of pandas, numpy, and scikit-learn for preprocessing, computation, and predictive modeling.

Use of Libraries

- pandas:
  - Imports CSV data into DataFrames.
  - Cleans missing or duplicate entries.
  - Groups data by session and computes average, maximum, and minimum angles.
  - Calculates Range of Motion (ROM) per session as ROM = max(angle) - min(angle).
- numpy:
  - Handles numerical operations such as moving averages and noise filtering.
  - Performs normalization and array-based mathematical computations.
  - Generates time-series arrays for regression modeling.
- scikit-learn:
  - Implements Linear Regression and Polynomial Regression models to identify recovery trends.
  - Calculates performance metrics such as R^2 Score, MAE, and MSE.
  - Supports predictive modeling for future ROM estimation based on past performance data.

The processed data is then plotted using matplotlib to generate session-wise progress graphs, showing both actual and predicted recovery curves.

10.4 Graphical User Interface (Tkinter GUI)

A Tkinter-based GUI was developed to create a user-friendly interface for physiotherapists and patients to interact with the analyzed data.

Key Features

- CSV Import: Allows users to select and load rehabilitation data files.
- Progress Visualization: Displays line and bar graphs representing session progress, ROM improvements, and prediction curves.
- Message Boxes and Alerts: Inform users of missing data, analysis completion, or errors.

- Automation Buttons: Provide quick access to key functions such as “Analyze Data,” “Predict Progress,” and “Export Report.”
- Interactive Dashboard: Combines Python analysis with an accessible interface, eliminating the need for command-line usage.

This GUI transforms the analytical component into a practical, non-technical interface suitable for clinical and educational environments.

10.5 Overall Software Workflow

1. ESP32 (Arduino) → Captures IMU data and uploads it to Firebase.
2. Firebase → Stores and organizes real-time motion data by session.
3. Python (pandas, numpy, scikit-learn) → Processes, analyzes, and models data.
4. Tkinter GUI → Presents results visually for user interpretation and decision-making.

This integration of embedded firmware, cloud infrastructure, and AI-driven analytics makes the Hand Paralysis Rehabilitation System a complete hardware–software ecosystem, capable of both performing physical therapy and evaluating recovery scientifically.

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11. WORKING PRINCIPLE

The **Hand Paralysis Rehabilitation System** operates through a carefully synchronized sequence of hardware and software interactions that enable both automated and patient-controlled exercises. The core principle of the system is to simulate natural hand and wrist movements using actuators during the **Training Phase**, and to capture patient-initiated motion data during the **Testing Phase**.

The following steps describe the working process in detail:

Step 1: Power-On and Initialization
When powered on, the system distributes energy to all components: ESP32, servo motors, pneumatic pump, solenoid valve, and MPU6050 IMU. The ESP32 runs its startup routine, while the IMU begins self-calibration. Status LEDs indicate operational readiness.

Step 2: System Setup and Wi-Fi Connection
The ESP32 configures GPIO pins, initializes I^2C communication with the IMU, calibrates sensors, and connects to Wi-Fi. Once online, it links to Firebase Realtime Database for automatic cloud logging.

Step 3: Mode Selection
A three-pin toggle switch sets the system phase:

- **Training Mode:** activates actuators for guided exercises.
- **Testing Mode:** deactivates actuators, relying on the IMU for voluntary motion capture.

The mode button cycles through three exercises:

1. Finger folding/unfolding
2. Wrist flexion/extension
3. Wrist ulnar/radial deviation

LEDs or console messages indicate the active mode.

Step 4: Training Phase (Assisted Motion)
In Training Mode, the ESP32 drives actuators to perform repetitive movements:

- **Finger exercise:** pneumatic bladder inflates/deflates for flexion/extension.
- **Wrist flexion/extension:** servos move wrist smoothly between angles.
- **Wrist deviation:** servos guide side-to-side motion within safe limits.

Movements maintain consistent speed, angle, and repetitions for effective rehabilitation.

Step 5: Testing Phase (Patient-Controlled Motion)
In Testing Mode, the patient moves voluntarily while the IMU captures:

- Angular position and velocity
- Motion stability (tremors or fluctuations)

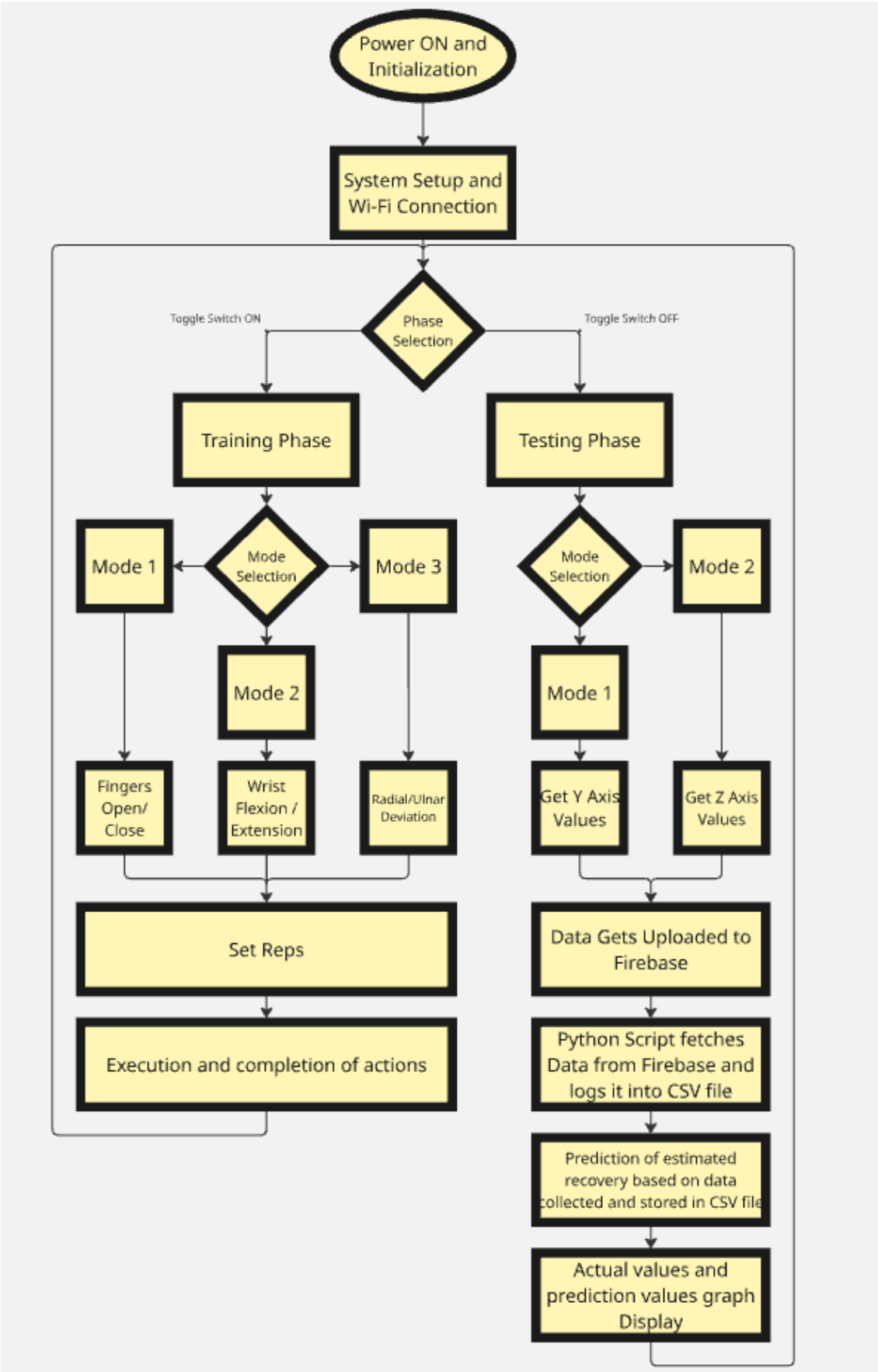
Data is recorded at regular intervals (e.g., 1 second) for accurate trend analysis.

Step 6: Data Logging and Cloud Transmission
ESP32 packages session ID, mode, repetition number, angle, and timestamp into JSON and uploads it to Firebase. Data is stored hierarchically for easy retrieval and visualization.

Step 7: Remote Monitoring and Analysis
Therapists access motion data via Firebase dashboards to:

- Track performance across sessions
- Compare assisted vs. voluntary motion
- Identify trends in range of motion and stability

This provides quantitative feedback to guide therapy adjustments and measure rehabilitation progress.


Power ON and Initialization
System Setup and Wi-Fi Connection
Phase Selection
Toggle Switch ON
Toggle Switch OFF
Training Phase
Testing Phase
Mode 1
Mode Selection
Mode 3
Mode Selection
Mode 2
Mode 2
Mode 1
Fingers Open/ Close
Wrist Flexion / Extension
Radial/Ulnar Deviation
Get Y Axis Values
Get Z Axis Values
Set Reps
Data Gets Uploaded to Firebase
Execution and completion of actions
Python Script fetches Data from Firebase and logs it into CSV file
Prediction of estimated recovery based on data collected and stored in CSV file
Actual values and prediction values graph Display

12. MATHEMATICAL AND ANALYTICAL MODELLING

This section explains the mathematical and analytical framework behind the Hand Paralysis Rehabilitation System, covering kinematic calculations, control algorithms, signal processing, and predictive modelling for patient recovery analysis.

10.1 Kinematics Calculations: MPU6050 Stabilization:
The MPU6050 sensor continuously measures wrist acceleration and angular velocity to track motion in real time. Using quaternion-based sensor fusion and filters such as the Mahony algorithm, the system eliminates drift and noise to provide stable wrist orientation tracking. This ensures accurate detection of wrist flexion, extension, and deviation during the training phase.

Quaternion rate from gyroscope:

$$\dot{q} = \frac{1}{2} q \otimes \omega_{raw} \text{with} \omega_{raw} = [0, G_x, G_y, G_z]^T$$

Discrete update (Euler):

$$q_t = q_{t-1} + \dot{q}\,\Delta t \text{then normalize } q_t$$

Accelerometer-based correction (Mahony-style):

$$e = g_{meas} \times g_{est}$$
$$\omega_{err} = K_P e + K_I \int e\,dt$$

Corrected angular rate applied in quaternion update:

$$\omega = \omega_{raw} + \omega_{err}$$

This fusion of gyroscope and accelerometer data provides smooth and drift-free wrist tracking throughout the exercise cycle.

10.2 Inverse Kinematics for Servo Positioning:
To accurately replicate wrist movements, the system employs **Inverse Kinematics (IK)** for servo motor control. The wrist mechanism is modelled as a **two-link planar arm**, where mathematical IK equations determine servo angles θ_1 and θ_2 for any desired wrist position.
Forward kinematics (end-effector at (x, y)):

$$x = L_1 \cos\,\theta_1 + L_2 \cos\,(\theta_1 + \theta_2)$$
$$y = L_1 \sin\,\theta_1 + L_2 \sin\,(\theta_1 + \theta_2)$$

Solving for joint angles:

$$\cos\,\theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$
$$\theta_2 = \text{atan2}(\pm\sqrt{1 - \cos^2\theta_2}, \cos\,\theta_2)$$
$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(L_2 \sin\,\theta_2, L_1 + L_2 \cos\,\theta_2)$$

This ensures smooth, human-like wrist motion and coordinated servo positioning during rehabilitation exercises.

10.3 Power & Motor Control Expansion

The control subsystem drives servo motors and pneumatic actuators for finger and wrist rehabilitation. A regulated power circuit ensures consistent torque, while the ESP32 synchronizes all actuators with precise timing.

Electrical power and torque:

$$P = V \cdot I, \tau = K_t I$$

DC motor speed approximation (neglecting dynamics):

$$\omega \approx K_v(V_{applied} - IR)$$

PWM duty-to-voltage mapping:

$$V_{applied} = D \cdot V_{supply} \text{(where } 0 \leq D \leq 1 \text{ is PWM duty)}$$

Servo position control (PID):

$$u(t) = K_P e(t) + K_I \int e(t)\, dt + K_D \frac{de(t)}{dt}$$
$$e(t) = \theta_{target} - \theta_{meas}$$

This ensures stable servo movement with minimal overshoot and accurate position control.

10.4 Advanced Signal Analysis for Progression

Raw IMU and servo feedback data are filtered using digital bandpass filters to remove high-frequency noise and ensure clean motion profiles for analysis.

Bandpass filter transfer function (Butterworth example):

$$H(s) = \frac{\omega_0^2}{s^2 + \sqrt{2}\omega_0 s + \omega_0^2}$$

Discrete form (using bilinear transform):

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] - a_1 y[n-1] - a_2 y[n-2]$$

Feature extraction for rehabilitation analysis:

$$ROM_{session} = \max(\theta(t)) - \min(\theta(t))$$
$$smooth\, ROM = moving_avg(ROM, w)$$

This filtering and feature extraction process enhances accuracy in detecting progress trends across multiple rehabilitation sessions.

10.5 Time-Series Predictive Modeling

To predict recovery progress, the system employs machine learning algorithms — Linear Regression, Polynomial Regression, and LSTM neural networks — implemented using Python.

Linear regression model:

$$\hat{y}_t = \beta_0 + \beta_1 t$$

Polynomial regression (degree d):

$$\hat{y}_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots + \beta_d t^d$$

LSTM cell operations (per time step t):

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$
$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$
$$\tilde{c}_t = \tanh\,(W_c x_t + U_c h_{t-1} + b_c)$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$
$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$
$$h_t = o_t \odot \tanh\,(c_t)$$
$$\hat{y}_{t+1} = Dense(h_t)$$

These models process time-series wrist motion data from Firebase to forecast recovery rates, identify improvement trends, and estimate the expected recovery timeline.

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13. DATA PROCESSING

The proposed system offers a **cost-effective, portable, and data-driven solution** for hand and wrist rehabilitation, significantly improving upon traditional therapeutic methods and existing commercial rehabilitation devices.
By combining real-time sensor data acquisition, intelligent cloud storage, and advanced data analytics, the system enables quantitative evaluation of patient progress. This not only provides measurable feedback to therapists but also allows adaptive therapy adjustments based on recovery trends over time. The following subsections describe how data is collected, processed, and analyzed for predictive rehabilitation outcomes.

11.1 Data Collection Process:
During the Testing Phase, the MPU6050 inertial measurement unit (IMU) continuously measures wrist motion across two axes — X (flexion–extension) and Y (ulnar–radial deviation) — to calculate the Range of Motion (ROM), a primary metric for assessing rehabilitation progress.
Each rehabilitation session is uniquely identified and logged through the ESP32 microcontroller, which transmits all relevant parameters to the Firebase Realtime Database in real time. The key data parameters include:

- Session ID: Unique identifier for each training or testing session.
- Mode Number: Indicates the type of motion performed (finger, flexion–extension, or deviation).
- Timestamp: Time of each recorded data point for session-wise tracking.
- Angular Motion Data: Captured wrist rotation angle in degrees over time.
- ROM Value: Computed as the difference between maximum and minimum angular displacement in each cycle.

To ensure accuracy and consistency, the data undergoes preprocessing steps such as calibration, outlier rejection, and moving-average filtering. These steps eliminate measurement noise caused by sensor drift, vibration, or involuntary tremors. The refined, high-quality dataset forms the foundation for predictive modelling and recovery analytics.
Additionally, the real-time cloud integration via Firebase enables seamless remote monitoring. Physiotherapists or researchers can visualize patient performance trends through live dashboards, ensuring data continuity across multiple sessions and therapy environments (home or clinic).

11.2 Data Analysis Process:
Once the filtered motion data is stored in Firebase, it is exported and analysed using Python-based analytical tools including pandas, NumPy, matplotlib, and scikit-learn. The primary goal of this analysis is to identify recovery patterns, quantify improvement, and predict future rehabilitation outcomes using statistical and machine learning models.

1. Regression-Based Trend Analysis:
Linear and Polynomial Regression models are applied to the session-wise ROM data to identify improvement trends over time.

- Linear Regression determines the overall rate of recovery through a straight-line fit.
- Polynomial Regression captures non-linear recovery curves, which more accurately represent real-world patient progress that accelerates after initial muscle reactivation.

These regression models provide fitted curves representing the relationship between session count and motion capability, allowing easy visualization of rehabilitation performance.

2. Time-Series Predictive Modelling (LSTM):
To model temporal dependencies in the data, Long Short-Term Memory (LSTM) neural networks are used. LSTMs can retain historical trends and predict future movement potential based on past performance.
This enables the system to estimate:

- The expected recovery timeline,
- The rate of improvement per session, and
- Predicted range of motion (ROM) for future therapy cycles.

3. Visualization and Performance Metrics:
The processed data is plotted as:

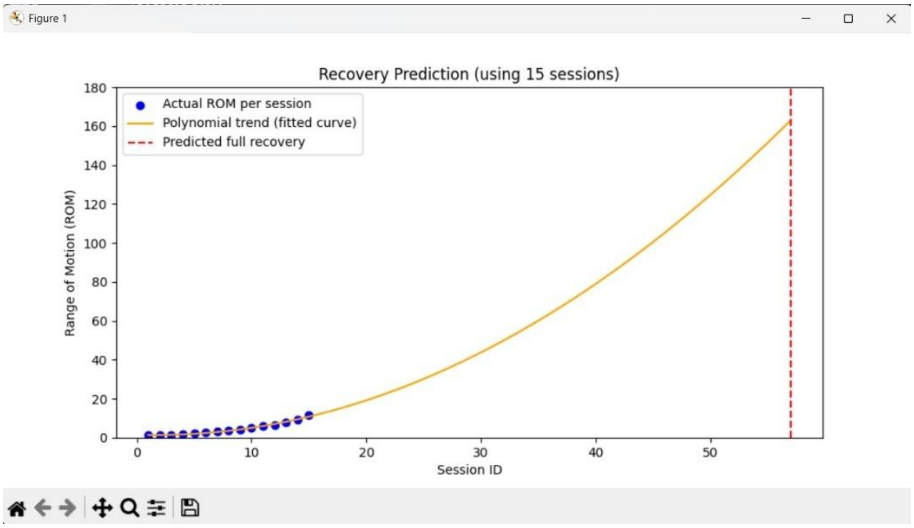
- Actual vs. Predicted ROM graphs, showing how well the models fit observed recovery data.
- Session progress charts, highlighting incremental improvements across multiple training sessions.

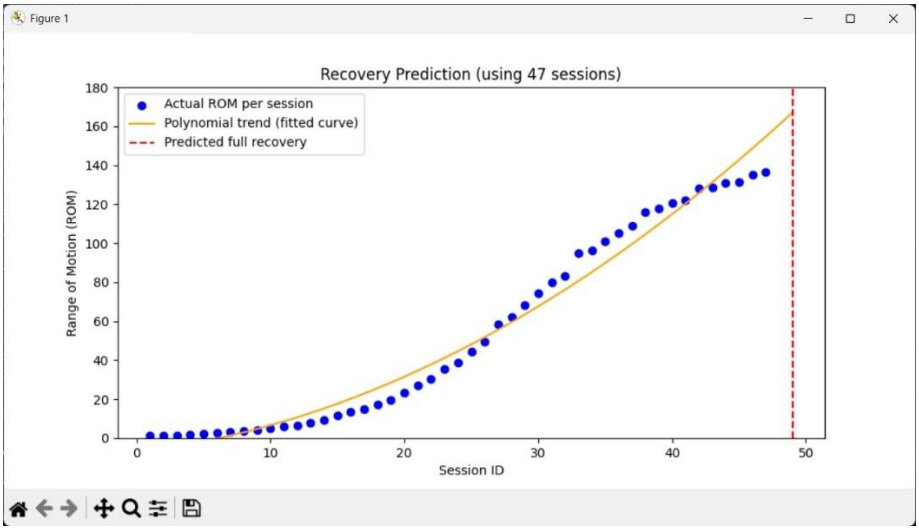
To evaluate the model's reliability, metrics such as Mean Squared Error (MSE) and Coefficient of Determination (R^2) are calculated.
Lower MSE and higher R^2 values indicate strong model accuracy and effective data correlation with patient performance.

4. Clinical Insight Generation:
The results are interpreted to provide quantitative insights for physiotherapists. Based on observed progress rates and predicted outcomes, the therapist can:

- Adjust training duration or motion difficulty,
- Reduce or increase machine assistance levels,
- Identify plateaus in recovery for targeted intervention.

Through this process, raw motion data is transformed into actionable intelligence, bridging the gap between engineering data and medical decision-making. This analytical framework ultimately supports personalized therapy planning, ensuring efficient and evidence-based patient rehabilitation.





14. EXPERT VALIDATION

The system was reviewed and validated by medical and rehabilitation experts under supervised testing. Their key feedback included:

- **Bionics Innovations (Certified Manufacturer):**
  Emphasized incorporating wrist twist motion to simulate natural hand behavior.
- **Dr. Sahab Thaver (Physiotherapist, Nanavati Hospital):**
  Suggested using the device for repetition-based therapy for sustained recovery.
- **Dr. Mansi Thakkar (Physiotherapist, Private Practitioner):**
  Recommended reducing assistance gradually to enhance patient engagement and muscle strength.

This expert validation confirms the system's functionality, therapeutic potential, and user-centered design.

15. ADVANTAGES

The proposed system offers a cost-effective, portable, and data-driven solution for hand and wrist rehabilitation, improving on traditional techniques and commercial devices.

1. **Multi-Axis Rehabilitation**
   - Supports both finger and wrist motion for holistic recovery.
   - Pneumatic module enables finger flexion/extension; servo-controlled assembly allows wrist flexion-extension and deviation.
   - Targets multiple muscle groups simultaneously, mimicking physiotherapist-guided exercises.
2. **Low-Cost, Portable Design**
   - Built with affordable, off-the-shelf components and 3D-printed parts.
   - Lightweight, compact, and battery-powered, suitable for home or remote use.
3. **Wireless Monitoring & Cloud Connectivity**
   - ESP32 uploads session data to Firebase in real time.
   - Enables remote progress tracking via web or mobile dashboards.
   - Reduces wiring, enhancing patient comfort and safety.
4. **Quantitative Progress Tracking**
   - IMU sensor records real-time angles, speed, and movement smoothness.
   - Enables data-driven therapy adjustments and long-term performance analysis.
5. **Safe & Controlled Operation**
   - Smooth motion profiles regulated pneumatic pressure, mechanical range limiters, and emergency stop ensure patient safety.
6. **Dual-Phase Adaptability**
   - **Training Phase:** Assisted movements for early-stage recovery.
   - **Testing Phase:** Active patient-driven movements for quantitative assessment.
   - Supports complete rehabilitation from assisted to independent motion.

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16. APPLICATIONS

The **Hand Paralysis Rehabilitation System** has a wide range of applications in the fields of **medical rehabilitation**, **physiotherapy**, and **biomedical research**.

Its flexible design allows it to be used across different recovery stages and environments - from clinical setups to home-based therapy systems.

1. Post-Stroke Hand and Wrist Rehabilitation

One of the primary applications of this system is in the **rehabilitation of stroke patients** who suffer from partial or complete paralysis of the hand and wrist. The device assists in restoring neuromuscular coordination by guiding the wrist and fingers through repetitive motion cycles. Such **repetitive, controlled exercises** are proven to enhance neural plasticity - the brain's ability to rewire motor pathways - thereby accelerating recovery of voluntary control.

2. Recovery After Nerve Injury or Orthopedic Surgery

Patients recovering from **peripheral nerve injuries**, **fractures**, or **tendon repairs** often experience stiffness and reduced mobility in their wrist or fingers. This rehabilitation system helps these patients by providing **safe, gradual joint mobilization** and **strength retraining** without the need for constant therapist supervision. It ensures that the healing tissues are moved within safe limits, preventing muscle atrophy and maintaining joint flexibility during recovery.

3. Physiotherapy Clinics and Rehabilitation Centers

The system can be used in **professional physiotherapy clinics** and **rehabilitation centers** as an **assistive training tool**. Therapists can configure the movement type and speed for each patient, while the cloud-based data logging allows them to **monitor progress remotely** or **store long-term patient data**. Its affordability and compact size make it suitable for **multi-patient use** or for integration into existing physiotherapy programs without major infrastructure changes.

4. At-Home Therapy for Paralyzed or Mobility-Impaired Patients

Because the device is **lightweight, low-cost, and Wi-Fi-enabled**, it can be conveniently used for **home-based rehabilitation**. Patients who cannot visit clinics regularly can perform daily exercises independently while therapists review their data remotely via the Firebase database. This greatly improves therapy accessibility and **ensures continuity of treatment**, which is crucial for long-term recovery outcomes.

5. Research in Robotic Rehabilitation and Human–Machine Interaction

Beyond clinical use, the project also serves as a **research platform** for studies in **robotic rehabilitation, biomechanics, and human–machine interaction**. Researchers can use the system to analyze motion patterns, evaluate control algorithms, or explore adaptive feedback mechanisms in assistive robotics. Its modular, open-ended design allows easy customization for testing new actuators, sensors, or AI-driven control methods in future rehabilitation technologies.

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17. FUTURE SCOPE

The **Hand Paralysis Rehabilitation System** presents significant potential for future enhancements through the integration of advanced technologies and smarter control methods. Several improvements can be made to further increase the system's efficiency, adaptability, and clinical usefulness.

1. Integration of Mobile App Dashboard for Real-Time Visualization
A dedicated **mobile or web application** can be developed to allow therapists and patients to **monitor real-time rehabilitation data** such as joint angles, repetition count, and session duration. Through an intuitive dashboard, the app can display **live motion graphs**, **session summaries**, and **progress analytics** in an easy-to-understand format. This would enable better engagement for patients during home therapy sessions and allow therapists to **track progress remotely** without accessing the raw Firebase data directly.

2. Use of Machine Learning Algorithms to Analyze Improvement Trends
Machine learning can be introduced to analyze the **large volume of motion data** collected during multiple therapy sessions. By training models on parameters like **angle range, speed, and smoothness**, the system can automatically **predict recovery trends**, detect **abnormal motion patterns**, and **recommend exercise adjustments**. Over time, the system could personalize the therapy routine based on the patient's improvement rate, creating a **data-driven adaptive rehabilitation model**.

3. Implementation of Adaptive Motion Profiles Based on Patient Progress
Currently, the movement patterns and speeds are preprogrammed and manually selected by the therapist or user. Future versions can incorporate **adaptive motion profiles** that automatically adjust movement speed, range, or resistance based on the patient's **measured strength and progress**. For instance, as the patient gains control, the system can gradually reduce assistance, promoting **active participation** and **muscle re-education**. This feature would make the device more **intelligent and responsive** to individual rehabilitation needs.

4. Incorporation of Voice or Gesture Control for Enhanced Accessibility
To improve **ease of use**, especially for patients with limited hand mobility, the system can be enhanced with **voice commands** or **gesture-based controls**. Using simple voice recognition modules or camera-based gesture detection, patients could start, stop, or change modes **without pressing physical buttons**. This would make the device more **accessible to individuals with severe paralysis or limited mobility**, while also improving overall **user independence and comfort**.

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18. CONCLUSION

The **Hand Paralysis Rehabilitation System** successfully combines hardware, control electronics, and cloud-based data logging into a unified rehabilitation platform. Unlike conventional systems that focus solely on finger motion, this device provides **multi-axis training**, covering both **finger and wrist movement**.

The inclusion of **Training and Testing phases** allows the patient to gradually transition from assisted motion to voluntary control. The integration of an **IMU sensor** and **Firebase cloud** ensures that each session's data is logged, enabling personalized progress tracking and data-driven medical assessment.

This project demonstrates how **embedded systems, mechanical actuation, and IoT technologies** can come together to create an affordable, intelligent rehabilitation solution that empowers patients to regain independence and functionality.

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